LFM Pulse Compression, Link Budget Modeling & +10 dB SNR Enhancement for Long-Range Radar Target Detection

Track 1: Radar Signal Processing | Curator: Mr. Rajendra Kumar (Scientist, Radar Division, SDSC SHAR)
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I. PROBLEM STATEMENT & MATHEMATICAL FORMULATION

A. Radar Dilemma: In long-range tracking radars (e.g., SDSC SHAR space launch tracking at R = 200 km, RCS = 1.0 m²), detecting weak echoes demands high pulse energy $E_t = P_t \cdot \tau_p$. For simple unmodulated pulses, range resolution degrades linearly with pulse duration: $\Delta R = c \cdot \tau_p / 2$. Widening an unmodulated pulse from 10 µs to 100 µs scales energy by +10 dB but severely corrupts resolution from 1.5 km to 15.0 km, destroying target discrimination.
B. Linear Frequency Modulation (LFM): Pulse compression resolves this dilemma by sweeping carrier frequency over bandwidth B = 4.0 MHz within pulse duration τ_p: $s(t) = \text{rect}(t/\tau_p) \exp(j \pi k_r t^2)$, with chirp slope $k_r = B / \tau_p$. The matched filter impulse response $h(t) = s^*(-t)$ yields compressed autocorrelation $y(t) = \tau_p \cdot \text{sinc}(\pi B t)$. Crucially, the compressed range resolution is governed *exclusively* by sweep bandwidth B, independent of pulse width τ_p:

$$\Delta R = c / (2B) = 3.0\times10\blacksquare / (2 \times 4.0\times10\blacksquare) = 37.50 \text{ m (Identical for } \tau = 10 \text{ µs and } \tau = 100 \text{ µs)}$$

C. Radar Link Budget & +10 dB SNR Scaling: Received target power P_r at R = 200 km (P_t = 250 kW, G = 40 dBi, λ = 0.1 m [S-band, 3.0 GHz], L = 3.0 dB, F = 2.5 dB) via the Radar Range Equation is P_r = -134.04 dBW (3.946×10⁻¹■ W). Noise spectral density is $N\blacksquare = k_B T\blacksquare F$ = -201.48 dBW/Hz (7.12×10⁻²¹ W/Hz). Matched filter peak output SNR is $SNR_{MF} = 2 E_r / N\blacksquare = P_r \tau_p / N\blacksquare$. Comparing $\tau\blacksquare$ = 100 µs against $\tau\blacksquare$ = 10 µs yields the fundamental +10 dB SNR boost:

$$\Delta SNR_{MF} = 10 \log\blacksquare\blacksquare(\tau\blacksquare / \tau\blacksquare) = 10 \log\blacksquare\blacksquare(100 \text{ µs} / 10 \text{ µs}) = +10.00 \text{ dB (Time-Bandwidth: 40 vs 400)}$$

II. PROPOSED METHODOLOGY & RADAR PROCESSING PIPELINE

The complete end-to-end discrete-time radar processing chain executes: (1) Dual LFM chirp generation (B = 4 MHz, $\tau\blacksquare$ = 10 µs, $\tau\blacksquare$ = 100 µs) at f_s = 16 MHz; (2) Pulse train synthesis at PRT = 2.0 ms (PRF = 500 Hz, R_{unamb} = 300 km); (3) Target channel modeling at R = 200 km (delay τ_d = 1.333 ms) with circular complex AWGN; (4) Zero-padded FFT cross-correlation matched filtering preventing circular wraparound; (5) Adaptive 1-D CA-CFAR detection (48 reference cells, 16 guard cells, $P_{fa} = 10\blacksquare\blacksquare$).

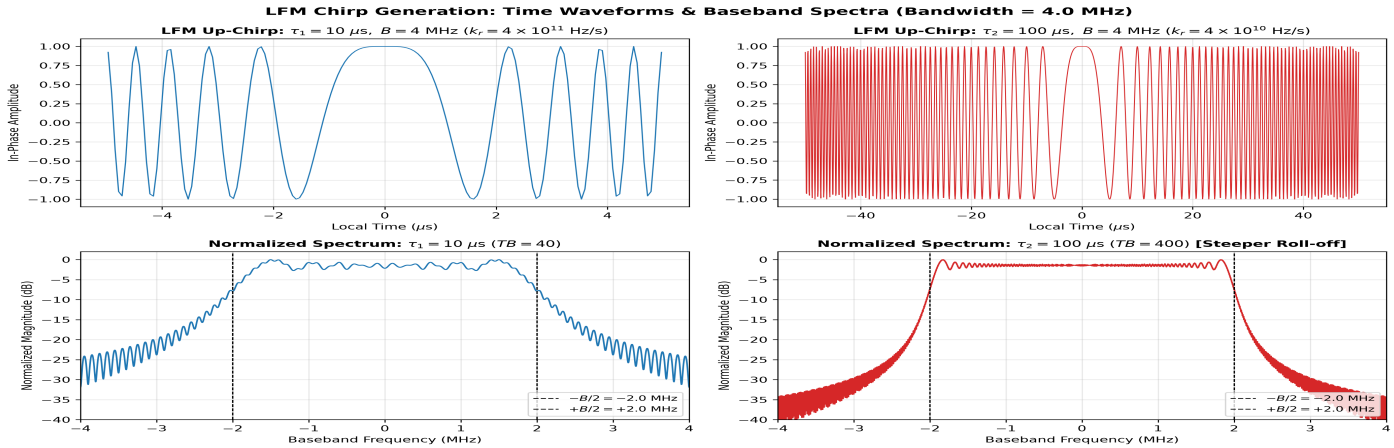


Fig. 1: LFM chirp time-domain waveforms and normalized frequency spectra (B = 4.0 MHz) for 10 µs and 100 µs pulses.

III. QUANTITATIVE RESULTS, BENCHMARK COMPARISON & VALIDATION

Radar System Metric / Parameter	Pulse 1 (τ■ = 10 μs)	Pulse 2 (τ■ = 100 μs)	Enhancement / Comparison
Pulse Duration (τ_p)	10.0 μs	100.0 μs	×10.0 (+10.0 dB)
Duty Cycle (τ_p / PRT, PRT = 2 ms)	0.50 %	5.00 %	×10.0
Radiated Pulse Energy (P_t = 250 kW)	2.50 J	25.00 J	+10.0 dB
Chirp Sweep Slope (k_r = B / τ_p)	4.00×10¹¹ Hz/s	4.00×10¹■ Hz/s	1/10 slope
Time-Bandwidth Product (TB = B · τ_p)	40.0 (16.02 dB)	400.0 (26.02 dB)	+10.0 dB compression gain
Theoretical Matched-Filter SNR	17.44 dB	27.44 dB	**+10.00 dB exact boost**
Simulated Measured Peak SNR	17.13 dB	27.31 dB	**+10.18 dB (Validates Theory)**
Nominal Range Resolution c/(2B)	37.50 m	37.50 m	**Identical Resolution (1.00×)**
Measured Half-Power (-3 dB) Width	35.36 m	34.17 m	**0.97× (Identical Discrimination)**
Peak Sidelobe Level (PSL)	-10.49 dB	-11.89 dB	Matches sampled discrete sinc
Required SNR (P_D = 0.90, P_fa = 10■■)	13.18 dB	13.18 dB	Marcum Q nonfluctuating model
Theoretical Link Margin	+4.25 dB	+14.25 dB	**+10.00 dB Link Margin Boost**
CA-CFAR Target Detection Margin	+6.81 dB	+16.99 dB	**+10.18 dB above adaptive thresh**

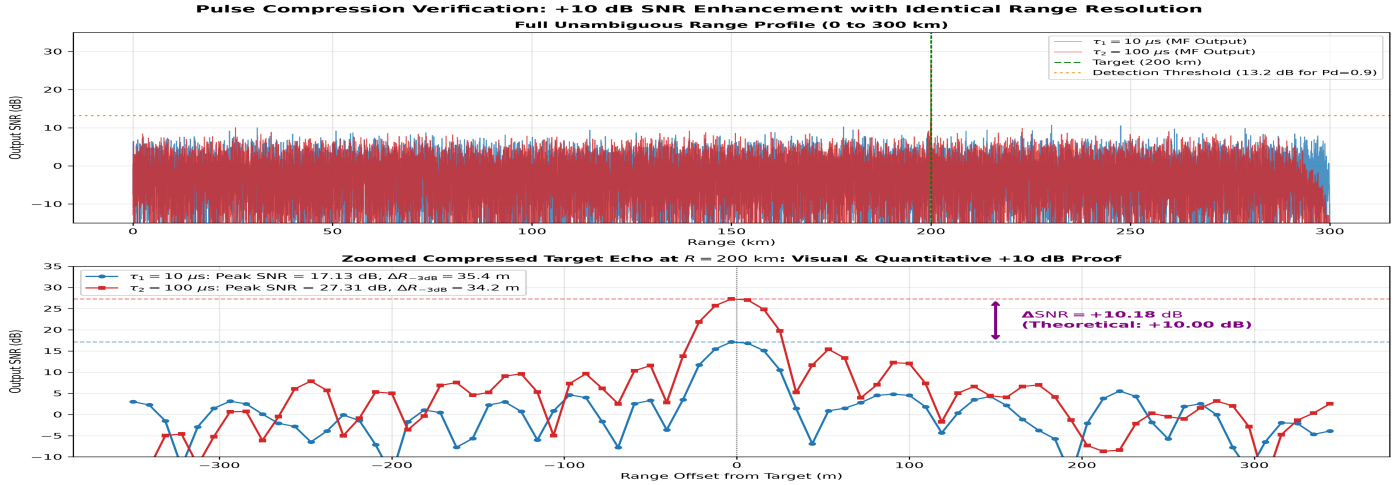


Fig. 2: Matched filter output at R = 200 km demonstrating the +10.18 dB SNR boost and identical 35 m mainlobe resolution.

IV. KEY TAKEAWAYS, PRACTICAL CONSIDERATIONS & LIMITATIONS

1. Operational Significance: While a 10 μs pulse yields a marginal link margin (+4.25 dB) vulnerable to rain fade and pointing errors, the 100 μs pulse provides an ironclad +14.25 dB link margin and +16.99 dB CFAR margin for mission-critical tracking at SDSC SHAR.
2. Minimum Range Blind Zone: In monostatic radar, transmission blinds reception over R_min = c τ_p / 2. For τ = 100 μs, R_min = 15.0 km (versus 1.5 km for 10 μs). Operational tracking radars employ dual-pulse waveforms: short pulses for close-range acquisition and long chirps for distant targets.
3. Doppler Shift Coupling: Radial velocity v_r induces Doppler frequency f_d = 2v_r / λ, causing a peak time shift Δt = -f_d / k_r. For space targets, alternating up- and down-chirps effectively resolves range-Doppler ambiguity.